

CO2 ASSESSMENT 3

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

A. Handling Missing Values

i. Missing Values Identified

- Age: C2
- Monthly_Spend: C3
- Purchase_Frequency: C5

ii. Two Imputation Techniques

1. Median Imputation
2. k-NN Imputation

iii. Imputed Values (Using Median)

- Age: Median = 33.5
- Monthly_Spend: Median = 41.5
- Purchase_Frequency: Median = 10

Justification: Median is preferred because it is less affected by extreme values such as **Age = 105** and **Monthly_Spend = 500**.

B. Outlier Detection and Treatment

i. Z-Score Method

Use the Z-score formula:

ii. Outliers Identified

- Age = 105 (C4)
- Monthly_Spend = 500 (C4)
- Monthly_Spend = 300 (C8)

iii. Treatment

- Apply **capping (winsorization)** or **remove** extreme values.
- **Justification:** Prevents outliers from affecting model accuracy.

C. Categorical Data Standardization

Inconsistent Gender Values

- M
- Male
- male
- F
- Female
- FEMALE

Standardized Format

- Male
- Female

Example:

Original	Standardized
M	Male
male	Male
FEMALE	Female
F	Female

Q2. Covariance & Data Reduction in E-commerce Analytics (*L4, L5*)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

A. Covariance Analysis

i. Covariance between Income and Purchase_Value

Using the given data:

Income (₹K)	Purchase Value (₹K)
40	100
60	150
85	210
120	320
45	110
75	180
150	360
70	160

Result: Covariance is **positive**.

ii. Interpretation

- As **Income increases**, **Purchase Value also increases**.
- Hence, the relationship is **positive**.

iii. Importance

- Identifies relationships between features.
- Helps detect redundant attributes.
- Improves feature selection for modeling.

B. Feature Selection

Selected Features

- Income
- Purchase_Value
- Credit_Score
- Orders

Justification

- These features strongly influence customer classification.
- Reduces redundancy and improves model efficiency.

C. Data Transformation

Standardization

Standardize all numerical attributes except **Cust_ID** and **Customer_Class**.

Why Standardization?

- Brings features to a common scale.
- Improves model accuracy.
- Prevents attributes with large values from dominating the analysis.

Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

A. Equal-Width Binning

Study Hours

- Range = $11 - 2 = 9$
- Bin Width = $9/3 = 3$

Bins:

- **Bin 1:** 2–4
- **Bin 2:** 5–7
- **Bin 3:** 8–11

Test Score

- Range = $100 - 40 = 60$
- Bin Width = $60/3 = 20$

Bins:

- **Bin 1:** 40–59
- **Bin 2:** 60–79
- **Bin 3:** 80–100

B. Equal-Frequency Binning

Study Hours

Bin 1: 2, 3, 4

Bin 2: 5, 6, 7

Bin 3: 8, 9, 10, 11

Test Score

Bin 1: 40, 50, 55

Bin 2: 65, 75, 85

Bin 3: 90, 95, 98, 100

C. Concept Hierarchy

Study Hours

- **Low:** 2–4
- **Medium:** 5–7
- **High:** 8–11

Test Score

- **Poor:** 40–59
- **Average:** 60–79
- **Excellent:** 80–100

Result: Discretization simplifies continuous data and improves interpretation for decision-making.

4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) Min-Max Normalization (Range 0 to 1)

Formula:

Here,

- **Min = 200**
- **Max = 1000**

Original Value

Normalized Value

200	0.00
300	0.125
400	0.25
600	0.50
1000	1.00

(b) Z-Score Normalization

Formula:

Where:

- **Mean (μ) = 500**
- **Standard Deviation (σ) $\approx$ 282.84**

Original Value	Z-Score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Result

- **Min-Max Normalization:** Scales data between **0 and 1**.
- **Z-Score Normalization:** Centers data around **mean = 0** with **standard deviation = 1**.

5. Data Transformation in Online Shopping Behaviour Analysis (L3, L4, L5)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent**

(in seconds) on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

III Sample Dataset: Visit Duration

Customer_ID Visit_Duration (seconds)

C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

A. Equal-Frequency Binning

Dataset

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

3 Bins

- **Bin 1:** 15, 30, 45, 60
- **Bin 2:** 120, 300, 600
- **Bin 3:** 900, 1800, 3600

Analysis: Equal-frequency binning groups customers into **short, medium, and long visit durations**.

B. Z-Score Normalization

Formula:

- Compute the **mean** and **standard deviation**.
- Convert each value to its **Z-score**.

Result: Z-score normalization standardizes the data and helps identify extreme values such as **1800** and **3600**.

C. Min-Max Normalization

Formula:

- **Min = 15**
- **Max = 3600**

Result: All values are scaled between **0 and 1**, making them suitable for clustering algorithms.

Comparison

- **Z-Score:** Better for handling outliers.
- **Min-Max:** Best for scaling data to the range **0–1**.